# Priyansh Vaish

Phone: +91 6351619265 | Email: priyansh.vaish01@gmail.com | LinkedIn | GitHub | Portfolio

## Professional Summary

AI/ML undergraduate focused on building and evaluating applied intelligent systems, from retrieval pipelines to automation-driven backend tools. Interned with FOSSEE, IIT Bombay, contributing open-source engineering work adopted across the platform. Now looking to bring that foundation to AI-focused engineering roles.

## Education

**VIT Bhopal University** — Bhopal, India
B.Tech in Computer Science (AI & ML) | CGPA: 8.68 — 2023 – 2027

## Technical Skills

**Languages:** Python, C++, SQL
**Generative AI:** Retrieval-Augmented Generation (RAG), LLM Integration
**Data & ML:** NumPy, Pandas, scikit-learn, Neural Networks, PyTorch
**Computer Vision:** OpenCV, YOLOv8, MediaPipe, Tesseract OCR, Depth Estimation, Image Processing
**Backend & Cloud:** FastAPI, Docker, Vercel, Supabase, Git
**Tools & Frameworks:** GitHub, n8n
**Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Computer Vision, Cloud Computing (AWS)

## Experience

**FOSSEE – OSDAG, IIT Bombay** — May 2026 – Jul 2026
Summer Intern — Mumbai, India

- Architected a client-side export/import engine for OSDAG's .3ps project format (JSON-based) using zlib and JSZip to compress and validate archives.
- Engineered carbon emission calculations across 4 categories – materials, transportation, machinery, and traffic diversion – applying NITI Aayog and SSP/RCP emission standards.
- Explored ML-based approaches to predict emission outcomes ahead of full calculation, as a potential optimization path for future iterations of the platform.

## Projects

**SkillTrace – Career Copilot** | GitHub — *Python |FastAPI |FAISS |RAG |LLM Applications |Transformer Embeddings*

- Built RAG pipeline matching resumes to job requirements: 95.8% retrieval accuracy across 6 resume profiles.
- Segmented retrieval limiting LLM context to top-3 chunks, cutting input tokens 40-60%.
- Orchestrated an asynchronous FastAPI backend integrating 3 LLM providers, reducing API response latency by 30% through concurrent request handling.

**FaceID Attendance System** | GitHub — *Python |OpenCV |face_recognition |Supabase*

- Optimized a real-time face recognition system (OpenCV): 90% accuracy, sub-200 ms per-frame latency.
- Developed an offline encoding pipeline with session-scoped deduplication, tested across 20+ encoded student profiles to validate automated batch processing as an alternative to manual attendance entry.
- Established 2-tier storage architecture (local CSV + Supabase PostgreSQL) for automated attendance tracking.

**EchoVision – Real-Time Edge Vision Assistant** | GitHub — *Python |Depth-Anything V2 |OpenCV |Piper TTS*

- Deployed Depth-Anything V2 (Small) for real-time hazard detection at 30 FPS, providing sub-200ms spatial feedback to assist visually impaired users in navigation.
- Devised a 3-region spatial engine (L/C/R) with Piper neural TTS and 15/30/45 degree turn guidance.
- Implemented modular Python architecture with 300ms cooldown logic to suppress duplicate real-time alerts.

## Achievements

**Finalist – Gnani.ai Buildathon, NASSCOM Tech Developer Hackathon 2025** — Oct 2025
Team Dikastirio | Track: Automation Using Agentic AI (Airlines Booking and Status)

- Delivered Hermes, a bilingual AI flight assistant with Twilio-integrated voice/SMS support for bookings, cancellations, and live status queries – finalist among 5,700+ registrations on the NASSCOM-HackerEarth platform.

## Certifications & Extracurriculars

- **AWS Certified AI Practitioner** – Amazon Web Services | Credential — Jul 2026
- **Machine Learning Specialization** – DeepLearning.AI & Stanford University | Credential — Sept 2025
- Design Team Member – AWS Cloud Club, VIT Bhopal University — 2024 – 2025